Smart Agri AI — Master Research & Defense Dossier

Project Scope: 24-Year Longitudinal APMC Price Forecasting Engine (1.31M Records, 358 Mandis, 8 Crops) | **Publication Target:** IEEE Access

EXECUTIVE DEFINITION: Smart Agri AI is a mathematically audited agricultural price forecasting and decision-support engine. Trained on **1,314,107 records across 358 APMC mandis (2001–2024)**, it utilizes a hierarchical market-partitioned **Random Forest Regressor (25 features)** evaluated through a **5-Fold Expanding Walk-Forward Cross-Validation** strategy across 13 years of out-of-sample forward horizons. It achieves a **Mean R^2 of 0.9489 ± 0.0118** and **MAPE of 7.84% ± 0.55%**, outperforming ARIMA (24.38%) and XGBoost (8.77%) while demonstrating **4× higher stability** with **zero data leakage**.

Section 1: The Exact 25 Engineered Features Specification

Target Variable: **modal_price** (Continuous exact price in Rs./Quintal). Features 12–22 are computed strictly within (Commodity, market) partitions to ensure cross_market_contamination == 0.

#	Feature Name	Category	Mathematical Formula / Definition	Domain Agricultural Rationale
1	**year**	Calendar	Calendar Year (2001–2024)	Captures multi-year macro inflation & long-term price drift
2	**month**	Calendar	Month of Year (1–12)	Identifies high-arrival harvest months vs lean supply months
3	**day**	Calendar	Day of Month (1–31)	Captures within-month auction liquidity and settlement cycles
4	**day_of_week**	Calendar	Day of Week (0=Mon, 6=Sun)	Captures weekly mandi closure days and auction volume peaks
5	**day_of_year**	Calendar	Julian Day of Year (1–366)	Models the continuous annual cyclical seasonal curve
6	**quarter**	Calendar	Fiscal Quarter (1–4)	Captures quarterly demand spikes (festivals, wedding seasons)
7	**is_kharif**	Seasonality	Binary flag: Harvest month $\in$ {Sep, Oct, Nov}	Autumn crop harvest surge (major monsoon crops: Soyabean, Rice)
8	**is_rabi**	Seasonality	Binary flag: Harvest month $\in$ {Mar, Apr, May}	Spring crop harvest surge (winter crops: Wheat, Gram)
9	**is_zaid**	Seasonality	Binary flag: Harvest month $\in$ {Jun, Jul, Aug}	Summer harvest supply gap dynamics and pre-monsoon volatility
10	**price_spread**	Market Struct.	max_price - min_price (Rs./Qtl)	Measures intra-day auction volatility and produce quality disparity
11	**log_arrivals**	Market Struct.	log(arrivals_tonnes + 1)	Supply volume elasticity: surge in arrivals drives downward price pressure
12	**lag_price_1**	Price Lag	Price at t - 1 day (within market & crop)	Immediate day-before price memory; primary short-term autoregressive feature
13	**lag_price_3**	Price Lag	Price at t - 3 days (within market & crop)	Captures short-term auction trend trajectory
14	**lag_price_7**	Price Lag	Price at t - 7 days (within market & crop)	Weekly cyclical price baseline (same weekday price benchmark)
15	**lag_price_14**	Price Lag	Price at t - 14 days (within market & crop)	Bi-weekly market momentum and trading cycle benchmark
16	**lag_price_30**	Price Lag	Price at t - 30 days (within market & crop)	Monthly price anchor for trend differentiation
17	**rolling_mean_7**	Rolling Stats	SMA_7(price) over past 7 active days	Filters high-frequency auction noise to reveal local moving baseline
18	**rolling_std_7**	Rolling Stats	StdDev_7(price) over past 7 active days	Quantifies localized market risk and short-term volatility
19	**rolling_mean_30**	Rolling Stats	SMA_30(price) over past 30 active days	Reflects the prevailing monthly regional equilibrium price
20	**ema_7**	Momentum	EMA_7 with smoothing factor alpha = 2/(7+1)	Fast-reacting price momentum indicator for sudden supply shocks
21	**ema_14**	Momentum	EMA_14 with smoothing factor alpha = 2/(14+1)	Medium-term price momentum indicator
22	**ema_30**	Momentum	EMA_30 with smoothing factor alpha = 2/(30+1)	Long-term trend filter distinguishing cyclical shifts from noise
23	**Commodity_encoded**	Categorical	Label Encoding ID (0 to 7)	Encodes biological & storage class (Wheat, Tomato, Onion, etc.)
24	**district_encoded**	Categorical	Label Encoding ID across Maharashtra	Captures district-level agro-climatic zone and logistics distance
25	**market_encoded**	Categorical	Label Encoding ID (0 to 357)	Captures APMC-specific physical market capacity and buyer density

Section 2: Research Gaps Filled by Smart Agri AI

Direct comparison against the four foundational reference works in agricultural AI literature:

Gap 1: Absence of a Trained ML Engine (IIT Patna Thesis — Kumar, May 2026)

The IIT Patna thesis developed a ReAct conversational chatbot but explicitly admitted on Page 37 & 47: *'the current prediction model uses moving average analysis [EMA-7/14/30] rather than a trained time-series forecasting model, limiting its ability to capture non-linear price dynamics.'*
How We Fill It: We replaced simple moving averages with a 25-feature non-linear Random Forest regressor trained on 1.31M records, achieving $R^2 = 0.9489$ and capturing seasonal harvest shifts, price spreads, and arrival volume dynamics.

Gap 2: Toy Classification vs Exact Price Regression (CMU / CoolCrop — Ma et al., 2018)

Ma et al. acknowledged on Page 7 that predicting exact prices is 'too difficult' and restricted their scope to 3-class directional movement (Up/Down/Same) over only 6 months of test data (Nov 2017–June 2018) across 3 markets, relying on synthetic SVD matrix completion.
How We Fill It: We solved the difficult continuous modal price regression problem (Rs./Qtl) over 24 years across 358 mandis, testing on 13 years of unseen future horizons without fabricating missing data via matrix completion.

Gap 3: Pure Conceptual Paper Without Measured Data (Zhinong AI — Li et al., May 2026)

Zhinong AI presented a conceptual design-science framework but explicitly conceded on Page 1 & 4: *'The study does not claim measured field performance because production logs and expert-labeled datasets were not available... all scores are analytic design ratings.'*
How We Fill It: Smart Agri AI provides hard, reproducible empirical machine learning science: 1,314,107 real observations, multi-model benchmarks, p-values, and an automated 8-point anti-leakage audit.

Gap 4: Pervasive Time-Series Data Leakage in Literature

Over 90% of published agri-forecasting papers use random K-Fold cross-validation (lookahead bias) and compute lag features globally, causing price shocks from one mandi to contaminate lags of another.
How We Fill It: We enforced hierarchical within-market grouping (cross_market_contamination == 0) and expanding walk-forward validation (zero future observations in training folds).

Gap 5: Macroeconomic Inflation Invariance Blindspot

Previous literature evaluates models over 6–12 months and assumes price stationarity, failing when macro-inflation shifts price baselines.
How We Fill It: We formulated the Inflation-Invariance Principle: proving why nominal MAE drifts with inflation while relative error (MAPE) remains constant at 7.84% ± 0.55% across 24 years.

Head-to-Head Literature Comparison Table

Dimension	CMU / CoolCrop (2018)	Zhinong AI (2026)	IIT Patna (2026)	Smart Agri AI (Ours)
Target Type	Direction (Up/Down/Same)	None (Conceptual)	Direction (EMA rule)	**Continuous Exact Price (Rs./Qtl)**
Dataset Scale	6 Months test / 3 markets	Zero empirical data	500 chatbot queries	**1,314,107 rows / 24 Yrs / 358 APMCs**
Core Algorithm	Nearest-Neighbor Classifier	Design framework	Rule-based Moving Avg	**Trained 25-Feature Random Forest**
Validation	Single static 6-mo split	None	User query testing	**5-Fold Expanding Walk-Forward**
Leakage Audit	Not reported (SVD impute)	N/A	Not tested	**8-Point Automated Audit (All PASS)**
Benchmark Comp.	Vs Logistic Regression	None	None	**Vs XGBoost (8.77%) & ARIMA (24.38%)**
Stability CV	Not evaluated	None	Not evaluated	**CV = 1.24% (R2), 4x more stable than XGB**

Section 3: Empirical Experimental Results & Baseline Comparison

All models evaluated on identical 5-fold expanding walk-forward splits across 1,314,107 records. Zero lookahead bias.

Model	Validation Strategy	Mean R^2	Std Dev R^2	Mean MAE	Mean MAPE	Stability (CV R^2)	Training Time
ARIMA (2,1,2)	Walk-Forward	0.7733	± 0.0941	Rs. 469.0	24.38%	12.17%	< 1 min
XGBoost Regressor	Walk-Forward	0.9113	± 0.0421	Rs. 198.4	8.77%	4.62%	~37 min
Random Forest (Ours)	**Walk-Forward**	**0.9489**	**± 0.0105**	**Rs. 156.0**	**7.84%**	**1.24%**	**~16 min**

5-Fold Detailed Breakdown (Proposed Random Forest)

Fold	Train Window	Val Window	Train Rows	Val Rows	R^2 Score	MAE (Rs.)	RMSE	MAPE (%)
Fold 1	2001–2010	2011–2013	349,869	177,370	**0.9354**	Rs. 145.84	304.86	**8.33%**
Fold 2	2001–2013	2014–2016	527,239	179,715	**0.9527**	Rs. 126.85	269.78	**7.90%**
Fold 3	2001–2016	2017–2018	706,954	181,787	**0.9660**	Rs. 111.22	239.50	**6.92%**
Fold 4	2001–2018	2019–2021	888,741	235,460	**0.9408**	Rs. 184.17	395.13	**8.16%**
Fold 5	2001–2021	2022–2024	1,124,201	189,906	**0.9495**	Rs. 211.68	470.56	**7.90%**
Mean ± Std	2001–Expanding	2011–2024	729,401 avg	192,848 avg	**0.9489 ± 0.012**	**Rs. 155.95**	**335.97**	**7.84% ± 0.55%**

Section 4: Automated 8-Point Anti-Leakage Verification Proof

Test ID & Check	Verification Method & Constraint	Status
T1: Target Exclusion	modal_price strictly excluded from feature matrix X (modal_price not in X.cols)	**PASS**
T2: Lag Backfill Integrity	Lag features never backfilled with contemporaneous or future target values	**PASS**
T3: Temporal Monotonicity	Zero backward date jumps within any APMC series (strictly sorted by date)	**PASS**
T4: Cross-Market Isolation	cross_market_contamination_count == 0 across all 358 APMC mandis	**PASS**
T5: Scaler Isolation	StandardScaler fit strictly on training fold; zero validation lookahead	**PASS**
T6: Holdout Integrity	Zero 2024 target observations present in training datasets	**PASS**
T7: AGMARKNET Isolation	Live operational records kept separate from historical ML training dataset	**PASS**
T8: Feature Parity	Training and runtime inference feature dimensions match identically (25 features)	**PASS**

Section 5: Viva / Presentation Defense Guide (What to Say)

Q: Why is Random Forest better than XGBoost here?
*A: While XGBoost achieves low error, it is sequentially gradient-boosted and overfits to regime shifts (demonetization, GST, COVID). Random Forest's bagged averaging produces **4x lower variance across folds (Std R^2 0.0105 vs 0.0421)**, making it far more reliable in production.*

Q: Why did MAE rise from Rs. 111 to Rs. 211 while MAPE stayed flat at ~7.9%?
A: Due to 24-year commodity price inflation. Base crop prices increased 250–350% (Soyabean rose from Rs. 1,800 to Rs. 4,500). An 8% error on Rs. 1,800 is Rs. 144, while 8% on Rs. 4,500 is Rs. 360. Constant MAPE proves the model learned market elasticity, not fixed prices.

Q: How does this beat the IIT Patna thesis?
A: IIT Patna's thesis explicitly stated in its limitations (p. 47) that its prediction was only an unlearned EMA moving average. We built the complete trained ML pipeline they identified as missing.